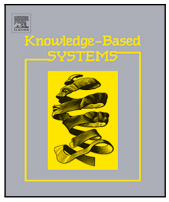
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RASNet: Recurrent aggregation neural network for safe and efficient drug recommendation

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ABSTRACT

Drug recommendation is one of the most crucial research topics in smart healthcare. Its goal is to provide a set of safe drug combination based on the patient's electronic health records (EHRs). Drug recommendation is challenging because it is difficult to obtain an appropriate representation of patient's health state from these personalized historical records. Meanwhile, drug recommendation must prioritize the safety of drug combination because drug–drug interactions (DDIs) could result in side effects. To address these issues, this paper proposes a novel recurrent aggregation neural network for safe drug recommendation, called RASNet. RASNet introduces a straightforward but efficient recurrent aggregation neural network to capture historical records related to the patient's health state of the current visit, which could improve the performance of EHR-based personalized modeling, particularly in cases where the patient's condition changes periodically. Furthermore, this paper presents a novel exponential controller for DDI loss to enhance the safety of drug combination. The proposed controller not only balances the DDI rate between the safety and accuracy of the drug recommendation but also ensures the performance even when the DDI rate is low. Extensive experiments on the MIMIC-III dataset demonstrate that RASNet achieves state-of-the-art performance. Moreover, RASNet exhibits excellent efficiency and safety in drug recommendation.

1. Introduction

AI-based drug recommendation is becoming increasingly crucial in the field of healthcare because of its significant research and practical value [1,2]. The main goal of drug recommendation is to customize a drug combination that is safe and effective for each patient based on his or her historical diagnoses [3–5]. At present, electronic health records (EHRs) offer invaluable opportunities for personalized drug recommendation. With large amounts of EHRs and drugs available, it can be challenging for healthcare professionals to stay updated with the latest and suitable treatments and medications [6]. Therefore, AI-based drug recommendation has the potential to offer valuable assistance to healthcare professionals in medical decision-making and enhance healthcare services [7].

Previous drug recommendation methods could be broadly categorized into rule-based [8], instance-based [9], and longitudinal methods [10,11]. Rule-based approaches heavily depend on the clinical expertise of medical professionals, which is time-consuming and prone to errors. Instance-based methods only consider the diagnosis during the current visit, disregarding the historical diagnosis information. Compared with the other two methods, longitudinal methods are superior in effectively modeling the patient's historical health condition [12]. Therefore, researchers have extensively used longitudinal methods for drug recommendation, which have gained significant popularity in recent years [13,14].

Although previous longitudinal methods have demonstrated promising results, it remains challenging when the patient's condition changes periodically. For example, the diagnosis of the current visit indicates anemia (285.21) caused by chronic kidney disease but it was not present in the previous visit. However, the same disease was also present during the visit before the last one, suggesting that the patient's condition may exhibit periodic similarity. Chronic diseases with periodic patterns may lead to inaccurate diagnoses on certain occasions, and models based on these diagnostic results need to consider the influence of this noisy data [15]. Therefore, the challenge is removing the noisy data from multiple diagnostic results and efficiently modeling the patient's historical diagnostic information.

Another challenge in drug recommendation is recommending a safe and side-effect-free drug combination. Patients with chronic conditions often require multiple medications for a long time, and various medications may cause adverse reactions. Drug safety is the primary concern of

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drug recommendation and significant efforts have been made to investigate the influence of DDIs on the recommended results. For example, Dual Molecular Graph Encoders for Recommending Effective and Safe Drug Combinations (SafeDrug) [10] introduced an adaptive controller to deal with DDI loss. This approach has a significant impact on the recommendation accuracy and effectively reduces the occurrence of DDI. Feature Fusion and Bipartite Decision Networks for Recommending Medication Combination (FFBDNet) [16] proposed a greedy loss to decrease the DDI rate to a safe level, but it led to a reduction in accuracy. Therefore, the primary challenge lies in ensuring the safety of the medication and the accuracy of its recommendation.

In response to these challenges, a novel drug recommendation model named RASNet is proposed in this paper. RASNet uses a recurrent aggregation network to identify historical health records that are similar to the current health status of the patient. It could filter out the noisy ones across multiple visits when a patient's condition exhibits periodic patterns, thereby enhancing the capability of patient representation. In addition, RASNet introduces a novel DDI loss to ensure the safety and accuracy of drug recommendation. The key contributions are summarized as follows:

- RASNet uses a recurrent aggregation network to incorporate historical health information that is similar to the patient's current health status. Therefore, RASNet could eliminate noisy data from the historical health information, enabling a more precise representation of the patient's health status.
- RASNet exploits the relationship between DDI loss and recommendation accuracy, and proposes an exponential controller to adjust the DDI loss. This approach could achieve a balance between the safety and accuracy of drug recommendation, and improve the overall performance.
- Extensive experiments show that RASNet not only exhibits good effectiveness and safety in drug recommendation but also has relatively low space and time complexity. The code is available on GitHub.[1]

2. Related work

With the advancement of deep learning in recent years, numerous studies have focused on leveraging this technique to enhance the precision of drug recommendation. The method used to derive vector representations of patients from their EHRs have a substantial impact on the effectiveness of drug recommendation. According to different representation learning methods of medical records, existing drug recommendation methods could be broadly classified into rule-based methods, instance-based methods, and longitudinal methods. Each method has unique pros and cons that play a crucial role in the development of drug recommendation.

2.1. Rule-based methods

Rule-based methods [8,17,18] depend on the insights and knowledge of clinical professionals. For instance, the Physician-Advisory System for CHF Management [19] used answer set programming to encode intricate clinical practice guidelines and produced treatment recommendation. Nevertheless, rule-based techniques require extensive involvement from clinical physicians and may exhibit limited adaptability in diverse scenarios.

2.2. Instance-based methods

Instance-based methods [20,21] only take the records of the current visit as input. However, these methods disregard the historical records if patients had several visits. For instance, Learning to Prescribe Effective and Safe Treatment Combinations for Multimorbidity (LEAP) [9] introduced a cost-effective and interpretable framework designed specifically for offering treatment recommendation to individuals with asthma. Meanwhile, Medical Knowledge Graph Embedding for Safe Medicine Recommendation (SMR) [22] embedded diseases, drugs, patients, and their corresponding relationships into a low-dimensional shared space and proposed to use link prediction to recommend drugs for patients.

2.3. Longitudinal methods

Longitudinal methods [23–26] fully leverage the patients' historical records and the temporal dependencies within these records. An Interpretable Predictive Model for Healthcare using Reverse Time Attention Mechanism (RETAIN) [27] used a dual-level recurrent neural network (RNN) with a reverse time attention mechanism, effectively establishing a longitudinal record model. Graph-attention Augmented Temporal Neural Network for Medication Recommendation (Gate) [28] used co-occurrence graphs to unveil correlations among clinical events and modeled structural relationships by introducing a graph attention mechanism, aiming to enhance the representations of patient medical records. In addition, Gate included a time-update module for learning the temporal dependencies among multiple hospital visits of the same patient. Multi-hop Reading on Memory Neural Network with Selective Coverage for Medication Recommendation (MRSC) [29] adopted a multi-hop memory neural network to store historical medical records and introduced a selective coverage mechanism to avoid excessive attention on the same record. SafeDrug [10] focused on modeling drug molecules and designed an adaptive loss controller to mitigate DDI, but it failed in effectively handling noisy data. Conditional Generation Net for Medication Recommendation (COGNet) [11] encoded medical records and drug information using transformers and introduced a novel copy or predict mechanism for drug generation. It achieved good results, but the significant increase in parameters led to low efficiency. Debiased, Longitudinal and Coordinated Drug Recommendation through Multi-Visit Clinic Records (DrugRec) [30] introduced the causal reasoning into the drug recommendation domain. It proposed a 2-SAT algorithm to harmonize DDIs and recommend safe drug combination for patients. A Drug Recommendation Model Based on Message Propagation and DDI Gating Mechanism (DRMP) [31] combined information diffusion neural networks and Bayesian neural networks. It achieved multiple recommendations and provided confidence scores for each recommendation. This assisted physicians in making more informed decisions. Symptom-based Set-to-set Small and Safe Drug Recommendation (4SDrug) [32] designed a set-oriented representation of drugs and symptoms and recommended drugs through set-to-set comparison to ensure that small and safe drug sets were recommended. FFBDNet [16] fused feature fusion and bipartite decision networks to integrate drug properties and patient records to improve predictive performance. Combinatorial Drug Recommendation with Substructure-aware Molecular Representation Learning (MoleRec) [33] delved into the importance of specific molecular substructures in drugs and designed an adjustment strategy to reweight the DDI loss in the training phase, enhancing the accuracy and safety of drug recommendation.

Existing drug recommendation methods have made progress, but several issues remain unresolved. First, the dynamic nature of chronic diseases requires a thoughtful examination of historical records to minimize the introduction of irrelevant information. Conversely, relying solely on the most recent medical visit record may result in an inaccurate representation of the patient. Second, there is a contradiction between the safety and accuracy of drug recommendation because a decrease in the DDI rate of the recommended drugs could lead to a decrease in recommendation accuracy, which indicates that safe drugs (low DDI rate) are always associated with inaccurate recommendations. To address these issues, this paper proposes a model called RASNet,

[1] https://github.com/cnhfeng/RASNet

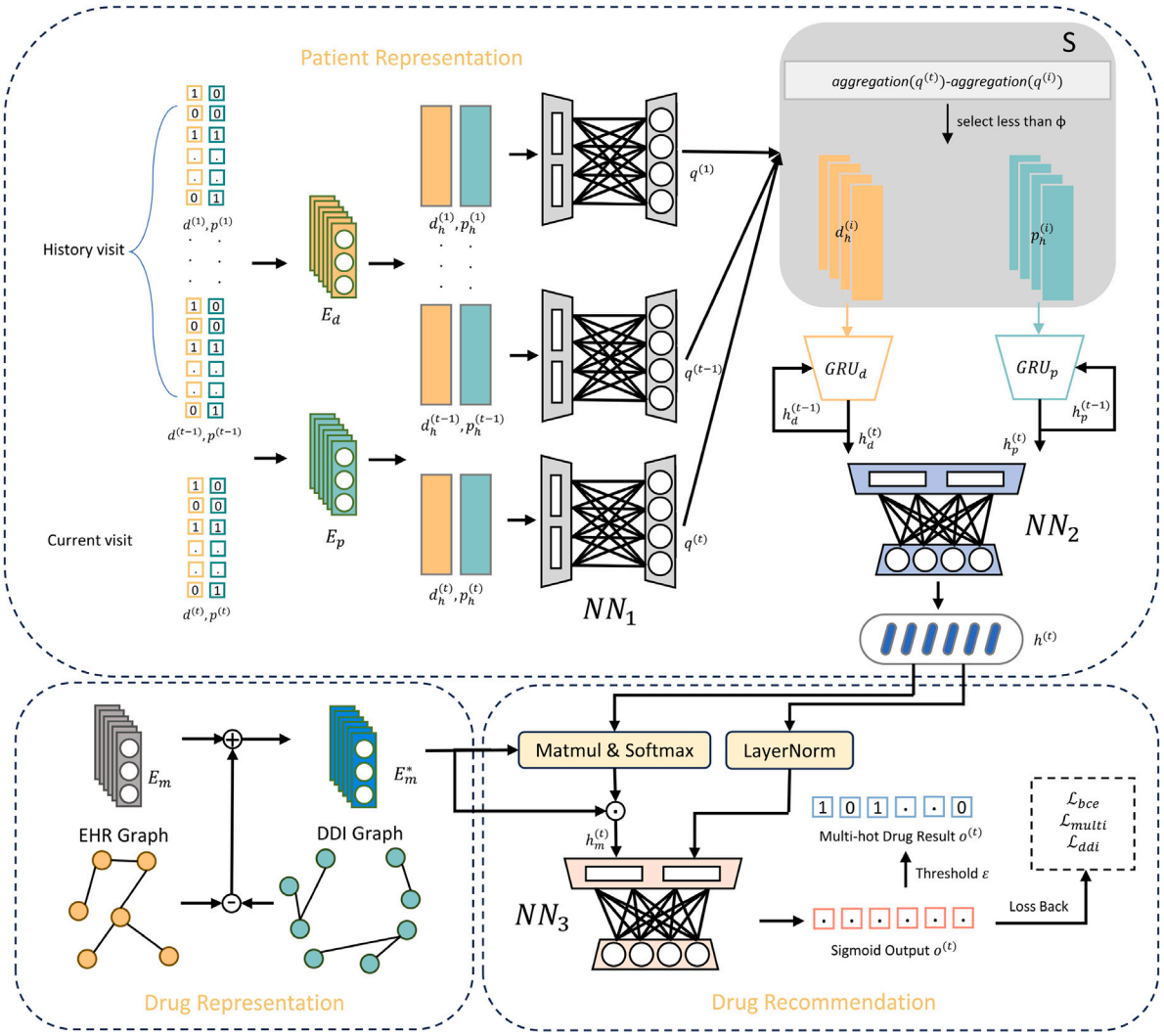


Fig. 1. The RASNet Model. RASNet includes three modules: patient representation module, drug representation module, and drug recommendation module. The patient representation module employs a recurrent aggregation neural network to effectively acquire and encode similar visit information. The drug representation module utilizes graph convolutional network to encode electronic health record and drug–drug interaction graph, resulting in safe drug representation. The drug recommendation module utilizes an attention mechanism to capture similarity information among drugs and integrates patient representations to provide drug recommendation.

which uses a recurrent aggregation neural network for safe drug recommendations. RASNet is designed to capture the health status in a manner similar to the current medical record. It effectively filters out noisy data that may arise when a patient's health condition undergoes periodic changes, thereby enhancing the representation of the patient's health status. Furthermore, RASNet presents an exponential controller to adjust DDI loss. The exponential controller could gradually reduce accuracy as the DDI rate decreases, thereby prioritizing drug safety and ensuring the recommendation accuracy.

3. Problem formulation

The model aims to predict the medication combination for the current patient visit based on their visit history. The objective is to increase accuracy by comparing the predicted combination with the ground truth provided by the doctor. Furthermore, it considers the safety factor, approximated using the DDI rate, which could potentially increase if two related drugs are recommended together.

3.1. EHR

The patient's EHRs consist of diagnosis, procedure, and medication information. Formally, patient j could be represented by a sequence $\mathbf{X}_j = [\boldsymbol{x}_j^{(1)}, \boldsymbol{x}_j^{(2)}, \ldots, \boldsymbol{x}_j^{(V_j)}]$, where V_j denotes the number of hospital visits for patient j. The ith visit of a patient could be expressed as $\boldsymbol{x}^{(t)} = [\boldsymbol{d}^{(t)}, \boldsymbol{p}^{(t)}, \boldsymbol{m}^{(t)}]$, where $\boldsymbol{d}^{(t)} \in \{0,1\}^{|\mathcal{D}|}$, $\boldsymbol{p}^{(t)} \in \{0,1\}^{|\mathcal{P}|}$, $\boldsymbol{m}^{(t)} \in \{0,1\}^{|\mathcal{M}|}$, with $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{M}$ representing the code sets of the diagnosis, procedure, and medication, respectively.

3.2. Drug recommendation

Formally, the adjacency matrix $\mathbf{A}_E$, $\mathbf{A}_D \in \mathbb{R}^{|\mathcal{M}|\times|\mathcal{M}|}$ is used to illustrate the construction of the EHR graph and DDI graph. $\mathbf{A}_E[i,j] = 1$ means that the ith and jth drugs have appeared in the same visit. For $\mathbf{A}_D$, only the pairwise DDIs are considered and $A_D[i,j] = 1$ means that the ith and jth drugs mutually interact. Given a patient's current diagnoses $\boldsymbol{d}^{(t)}$, procedures $\boldsymbol{p}^{(t)}$, historical visit information $[\boldsymbol{x}_j^{(1)}, \boldsymbol{x}_j^{(2)}, \ldots, \boldsymbol{x}_j^{(t-1)}]$, and the EHR and DDI adjacency matrix $\mathbf{A}_E$ and $\mathbf{A}_D$, the goal of the model is to recommend the safe drug combination $\hat{\boldsymbol{o}}^{(t)} \in \{0,1\}^{|\mathcal{M}|}$.

4. RASNet model

As shown in Fig. 1, RASNet comprises three key modules: (i) Patient representation module: this module is dedicated to effectively encoding

patient information. Initially, the crucial features of a patient undergo transformation into meaningful representation using embedding tables. Subsequently, a recurrent aggregation approach is used to encode a selection of the most analogous past visit records. By discerning differences among various visit records, this module captures essential characteristics from a patient's historical medical visits, thereby enhancing the understanding of the patient's health status. (ii) Drug representation module: this component is responsible for encoding distinct drug structures using graph convolutional networks (GCNs). (iii) Drug recommendation module: it uses an attention mechanism to acquire similar drug information. In collaboration with patient representations, it provides recommendations for drug combinations tailored to the patient's needs.

4.1. Patient representation module

This paper will use two embedding representations, specifically denoted as $\boldsymbol{d}_h^{(t)}$ and $\boldsymbol{p}_h^{(t)} \in \mathbb{R}^d$, to represent the patient's current health status. Nevertheless, relying solely on the current health condition is inadequate for making treatment decisions. In practical scenarios, physicians also require access to the patient's medical history before making prescriptions.

4.1.1. Diagnosis embedding

To make full use of longitudinal data in EHRs, this paper designs the diagnosis embedding table $\mathbf{E}_d \in \mathbb{R}^{|D| \times d}$. Each row of $\mathbf{E}_d$ represents a specific disease. There are $|D|$ diagnosis words $d_i (i \in \{1, 2, \ldots, |D|\})$ in the dataset. This paper sets d-dimensional random parameters for each diagnosis word and splices them together to get the diagnosis embedding table $\mathbf{E}_d$. The method used in generating the following procedure embedding table $\mathbf{E}_p$ and drug embedding table $\mathbf{E}_m$ is similar. The corresponding diagnosis codes $\boldsymbol{d}^{(t)} \in \{0, 1\}^{|D|}$ are projected into the embedding space:

$$\boldsymbol{d}_h^{(t)} = \boldsymbol{d}^{(t)} \mathbf{E}_d. \tag{1}$$

4.1.2. Procedure embedding

The procedure embedding table is denoted as $\mathbf{E}_p \in \mathbb{R}^{|\mathcal{P}| \times d}$. The corresponding procedure codes $\boldsymbol{p}^{(t)} \in \{0, 1\}^{|\mathcal{P}|}$ are projected into the embedding space:

$$\boldsymbol{p}_h^{(t)} = \boldsymbol{p}^{(t)} \mathbf{E}_p. \tag{2}$$

4.1.3. Recurrent aggregation

Previous studies [10,24] have considered the status change in a patient's health condition based on the most recent visit or all previous visits. However, they will become less efficient to model the situation when the patient's condition changes periodically. For example, the diagnosis of the current visit indicates anemia (285.21) caused by chronic kidney disease but it was not present in the previous visit. However, the same disease was present during the visit before the last one, suggesting that the patient's condition may exhibit periodic similarity. Chronic diseases with periodic patterns may lead to inaccurate diagnoses on certain occasions, and models based on these diagnostic results need to consider the influence of this noisy data. The challenge is how to automatically filter out the noisy data from multiple diagnostic results and efficiently model the patient's historical diagnostic information. To address this limitation, a recurrent aggregation network is designed to eliminate irrelevant information from the patient's history visits. It could improve the characterization of patient's health status by aggregating visit information, comparing it with the current visit, and filtering out irrelevant visits.

Specifically, RASNet concatenates the diagnosis and procedure embeddings and passes them through a neural network layer to derive the patient's health representation.

$$\boldsymbol{q}^{(t)} = NN_1(concat[\boldsymbol{d}_h^{(t)}, \boldsymbol{p}_h^{(t)}]), \tag{3}$$

where $\boldsymbol{q}^{(t)} \in \mathbb{R}^d$ represents the patient's health representation and $NN_1 : \mathbb{R}^{2d} \to \mathbb{R}^d$ represents a single-layer neural network. Subsequently, an aggregation function is designed to obtain the health status for each visit by aggregating the health representation from each visit. Then, a difference operation is used to quantify the change between the current and historical health representations. The specific implementation is as follows:

$$aggregation(\boldsymbol{q}^{(t)}) = Sigmoid(\mathbf{W}_2 ReLU(\mathbf{W}_1 \boldsymbol{q}^{(t)} + \boldsymbol{b}_1) + b_2), \tag{4}$$

$$s^{(i)} = \|aggregation(\boldsymbol{q}^{(t)}) - aggregation(\boldsymbol{q}^{(i)})\|, i \le t, \tag{5}$$

where $aggregation(\cdot)$ represents the aggregation function and $\mathbf{W}_1 \in \mathbb{R}^{d/2 \times d}$, $\boldsymbol{b}_1 \in \mathbb{R}^{d/2}$, $\mathbf{W}_2 \in \mathbb{R}^{d/2}$, and $b_2 \in \mathbb{R}$ are all learnable parameters. $s^{(i)} \in \mathbb{R}$ represents the degree of change between the current health status and the status at the ith visit. In this context, a hyperparameter φ is introduced. Visit changes $s^{(i)}$ less than φ result in the restoration of the embedding representations $d_h^{(i)}$ and $p_h^{(i)}$ in visit memory $\mathbf{S}$. The purpose of visit memory $\mathbf{S}$ is to store past visit information that is similar to the patient's current health status. In addition, the operation $\|\cdot\|$ involves taking the absolute value, which helps in minimizing the impact of excessively negative values. Therefore, the historical health condition $q^{(i)}$ within the visit memory will be highly similar to the current health condition $q^{(t)}$. Subsequently, two GRU units are used to capture sequential dependencies within the visit memory, thereby improving the accuracy of the patient's health representation.

$$\boldsymbol{h}_d^{(t)} = GRU_d([\boldsymbol{d}_h^{(i)}]^{\mathbf{S}}), \boldsymbol{d}_h^{(i)} \in \mathbf{S}, \tag{6}$$

$$\boldsymbol{h}_p^{(t)} = GRU_p([\boldsymbol{p}_h^{(i)}]^{\mathbf{S}}), \boldsymbol{p}_h^{(i)} \in \mathbf{S}, \tag{7}$$

where $\boldsymbol{h}_d^{(t)}$, $\boldsymbol{h}_p^{(t)} \in \mathbb{R}^d$ represent the hidden representations of the patient's diagnosis and procedure, respectively, and $\mathbf{S}$ is the visit memory. Finally, these two sets of hidden representations are concatenated, and the combined representation is passed through a feed-forward neural network denoted as $NN_2 : \mathbb{R}^{2d} \to \mathbb{R}^d$ to obtain the overall patient representation $\boldsymbol{h}^{(t)} \in \mathbb{R}^d$.

$$\boldsymbol{h}^{(t)} = NN_2(concat[\boldsymbol{h}_d^{(t)}, \boldsymbol{h}_p^{(t)}]), \tag{8}$$

4.2. Drug representation module

It is crucial for drug recommendation to recommend safe drug combinations, which indicates that DDI should be considered. Individuals suffering from chronic, long-term illnesses often require a diverse range of drugs, and DDIs may lead to various adverse effects. Therefore, RASNet proposes a joint modeling approach to effectively and safely recommend medications by considering both the patient's longitudinal records, represented as an EHR graph, and the drug knowledge base, represented as a DDI graph. The EHR graph shows drug associations prescribed during the same visit. The DDI graph provides DDI information, which may result in adverse reactions, and these drugs should not be taken simultaneously. Inspired by GAMENet [34], this paper uses GCNs to jointly encode the EHR and DDI graphs.

Given input $\mathbf{X} \in \mathbb{R}^{|\mathcal{M}| \times d}$ and the drug graph adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$, $GCN(\cdot, \cdot)$ is used to obtain drug representations.

$$GCN(\mathbf{X}, \mathbf{A}) = \sigma(\hat{\mathbf{D}}^{-\frac{1}{2}} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X}), \tag{9}$$

where $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$. $\mathbf{I} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$ is the identity matrix. $\hat{\mathbf{D}} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$ is the diagonal node degree matrix of $\hat{\mathbf{A}}$ ($\hat{\mathbf{D}}_{ii} = \Sigma_j \hat{\mathbf{A}}_{ij}$). Two layers of GCN are used to model drug interactions in both the EHR and DDI graphs. For the adjacency matrices $\mathbf{A}_E$ and $\mathbf{A}_D$ of the EHR and DDI graphs, respectively:

$$\mathbf{G}_e = GCN(ReLU(GCN(\mathbf{E}_m, \mathbf{A}_E))\mathbf{W}_3, \mathbf{A}_E), \tag{10}$$

$$\mathbf{G}_d = GCN(ReLU(GCN(\mathbf{E}_m, \mathbf{A}_D))\mathbf{W}_4, \mathbf{A}_D), \tag{11}$$

where $\mathbf{G}_e$, $\mathbf{G}_d \in \mathbb{R}^{|\mathcal{M}|\times d}$ represent the generated drug embeddings, $\mathbf{E}_m \in \mathbb{R}^{|\mathcal{M}|\times d}$ is the medication embedding table, and $\mathbf{W}_3$, $\mathbf{W}_4 \in \mathbb{R}^{d\times d}$ are learnable parameters. Finally, the medication embedding table $\mathbf{E}_m$ is combined with the two generated interaction representations $\mathbf{G}_e$ and $\mathbf{G}_d$ to obtain the final drug interaction:

$$\mathbf{E}_m^* = \mathbf{E}_m + (\mathbf{G}_e - \theta\mathbf{G}_d), \tag{12}$$

where θ is a learnable parameter.

4.3. Drug recommendation module

Similar to GAMENet [34], this paper uses an attention mechanism to calculate the similarity between patient representation $\boldsymbol{h}^{(t)}$ and drug representation $\mathbf{E}_m^* \in \mathbb{R}^{|\mathcal{M}|\times d}$, as shown below:

$$\boldsymbol{h}_m^{(t)} = \mathbf{E}_m^{*\top} Softmax(\mathbf{E}_m^* \boldsymbol{h}^{(t)\top}), \tag{13}$$

where $\boldsymbol{h}_m^{(t)} \in \mathbb{R}^d$ denotes the similarity of the drug information.

Finally, the drug recommendation is treated as a multi-label classification task. The probability of each drug being recommended could be obtained through the patient's hidden representation $\boldsymbol{h}^{(t)}$ and the similar drug information $\boldsymbol{h}_m^{(t)}$ as inputs to a neural network model. Formally, this is represented as follows:

$$\boldsymbol{o}^{(t)} = Sigmoid(NN_3(concat[LayerNorm(\boldsymbol{h}^{(t)}), \boldsymbol{h}_m^{(t)}])), \tag{14}$$

where $\boldsymbol{o}^{(t)} \in \mathbb{R}^{|\mathcal{M}|}$ represents the probability of the recommended drug combination and $NN_3 : \mathbb{R}^{2d} \to \mathbb{R}^{|\mathcal{M}|}$ is a single-layer neural network. $LayerNorm(\cdot)$ is the layer normalization function. Finally, a threshold $\varepsilon = 0.5$ is defined and items in the results that are greater than ε are selected as the recommended drugs.

4.4. Model training and inference

In the training phase, the model needs to optimize all parameters $\mathbf{W}_*, \boldsymbol{b}_*$ in the network to recommend drug combinations more accurately and safely.

- **Multi-label prediction loss.** RASNet uses two common loss functions for multi-label classification task in this model: binary cross-entropy loss $\mathcal{L}_{bce}$ and multi-label margin loss $\mathcal{L}_{multi}$. The binary cross-entropy loss encourages the model's predicted results to be close to the ground truth, whereas the multi-label margin loss ensures that the predicted probability of the ground truth labels has at least one margin larger than the others.

$$\mathcal{L}_{bce} = -\sum_{i=1}^{|\mathcal{M}|} m_i^{(t)} log(o_i^{(t)}) + (1 - m_i^{(t)}) log(1 - o_i^{(t)}), \tag{15}$$

$$\mathcal{L}_{multi} = \sum_{i,j:m_i^{(t)}=1,m_j^{(t)}=0} \frac{max(0, 1 - (o_i^{(t)} - o_j^{(t)}))}{|\mathcal{M}|}, \tag{16}$$

  where $o_i^{(t)}$ refers to the recommended drug i at the tth visit.
- **DDI loss.** The DDI loss is defined as follows:

$$\mathcal{L}_{DDI} = \sum_{i=1}^{|\mathcal{M}|}\sum_{j=1}^{|\mathcal{M}|} \mathbf{A}_{Dij} \cdot o_i^{(t)} \cdot o_j^{(t)}, \tag{17}$$

  where $\mathbf{A}_D \in \mathbb{R}^{|\mathcal{M}|\times|\mathcal{M}|}$ denotes the DDI adjacency matrix and $\cdot$ denotes the scalar product.

4.4.1. Overall controllable loss

The overall loss function is defined as follows:

$$\mathcal{L} = \alpha\mathcal{L}_{bce} + (1 - \alpha)\mathcal{L}_{multi} + \beta\mathcal{L}_{DDI}, \tag{18}$$

where α is a predefined hyperparameter and β is an exponential controller. The overall loss $\mathcal{L}$ includes the prediction loss (L_{bce} and $\mathcal{L}_{multi}$) and DDI loss ($\mathcal{L}_{DDI}$), which means that the prediction and DDI will affect the overall performance. Unlike previous methods [10,32,33], RASNet does not add mixing weights before the multi-label prediction loss because the high DDI rate of the drug combination will make the multi-label prediction loss larger. When the DDI of the drug combination is large, the DDI loss will increase accordingly. This situation may be caused by the inaccurate prediction of the model. Therefore, RASNet does not include a bias coefficient before the multi-label prediction loss to maintain the model's predictive capability.

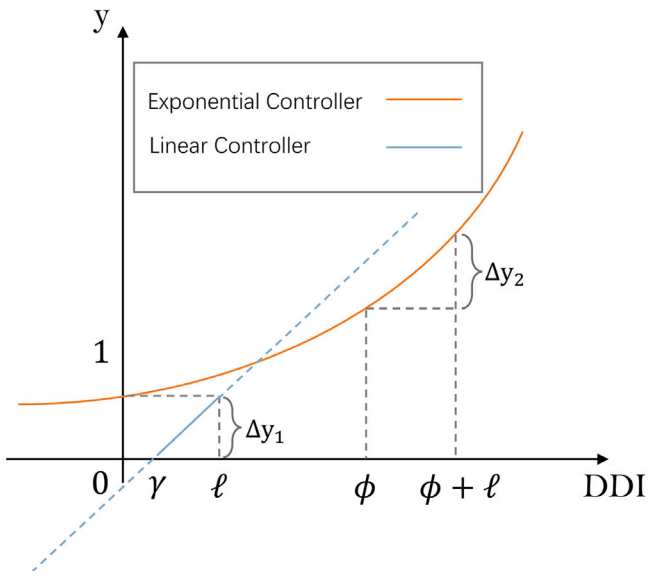


Fig. 2. Exponential Controller vs. Linear Controller. The DDI loss weight is designed as an exponential controller. Compared to a linear controller, the advantage of this design is that when DDI is small, the DDI loss increases slowly, which focuses on both the safety and accuracy of drug recommendation. However, when DDI is large, the DDI loss increases rapidly and with a larger range, thereby playing more emphasis on the safety of drug recommendation.

4.4.2. Exponential controller

Because the DDI rate and accuracy typically move in the same direction, as shown in previous work [10], a superior model should strive to maintain a low DDI rate while concurrently achieving a high level of recommendation accuracy. To achieve this balance between accuracy and safety in drug recommendation, this paper introduces an exponential controller denoted as β, which adaptively adjusts the loss term $\mathcal{L}_{DDI}$. Inspired by previous work [10], it is obvious that the variation in DDI exhibits an exponential relationship with other evaluation metrics. As a result, parameter β is designed as follows:

$$\beta = \delta^{DDI+\phi}, \tag{19}$$

where δ is the base of the exponent and ϕ is a predefined hyperparameter used to control the exponential controller β of the DDI loss. DDI represents the DDI in the model's recommended drug combination. The function here is not the linear controller in SafeDrug. As shown in Fig. 2, Δy_1 represents the weight range of the DDI loss in SafeDrug and Δy_2 represents the weight range of the exponential controller. Considering that cases of prescribing incorrect drugs are relatively rare in real-world data, this paper has defined an upper limit ℓ for DDI. When the DDIs rate in the predicted drug combination are low, β will slowly increase. In this scenario, RASNet accounts for both multi-label prediction and DDI loss whereas SafeDrug and MoleRec solely address the multi-label prediction loss. As the DDI loss of a drug combination increases, the rate of change of β will also increase, leading to a wider range of values. This adjustment enhances the safety of the drug recommendation. By contrast, SafeDrug and MoleRec elevate the DDI loss, leading to a reduction in multi-label prediction loss, thereby affecting

Algorithm 1 One training epoch of RASNet

Input: Training set $[X_j]_{j=1}^{N}$, EHR Graph A_E, DDI Graph A_D, hyper-parameters α, threshold φ and ε

for *patient* j to N **do**
  Read the patient j's history $x^{(1)}$, $x^{(2)}$, ... , $x^{(V_j)}$;
  for *history visit* $t = 1$ to V_j **do**
    Read the historical diagnoses, procedures of the patient at the t-th visit $[d^{(1)}, d^{(2)}, ..., d^{(t)}], [p^{(1)}, p^{(2)}, ..., p^{(t)}]$;
    Generate embeddings $d_h^{(i)}$ and $p_h^{(i)} (i \le t)$ by Eqs. (1)–(2);
    Generate degree of change $s^{(i)}$ by Eqs. (3)–(5);
    Generate patient representation $h^{(t)}$ by Eqs. (6)–(8);
    Generate drug interaction E_m^* by Eqs. (9)–(12);
    Generate similarity drug information $h_m^{(t)}$ by Eq. (13);
    Generate mapping representation $o^{(t)}$ by Eq. (14);
    Generate the output multi-hot drug vector $\hat{o}^{(t)}$ by comparing the matching scores $o^{(t)}$ with ε;
  end for
  Generate and accumulate $\mathcal{L}_{bce}$;$\mathcal{L}_{multi}$;$\mathcal{L}_{DDI}$ in Eqs. (15)–(17), respectively;
end for
Optimize the combined loss $\mathcal{L}$ in Eqs. (18)–(19);

Table 1
Data statistics.

Item	Size
# of visits/# of patients	15032/6530
diag./prod./med. space size	1958/1430/131
avg./max # of visits	2.37/29
avg./max # of diagnoses per visit	10.51/128
avg./max # of procedures per visit	3.84/50
avg./max # of medications per visit	11.44/65
total # of DDI pairs	448

the recommendation accuracy. By adjusting the hyperparameter ϕ, the exponential controller could more flexibly control the weight of DDI loss. This flexibility helps in balancing safety and accuracy, leading to improved performance.

The training algorithm is described in detail in Algorithm 1.

5. Experiments

5.1. Dataset

The Medical Information Mark for Intensive Care (MIMIC-III) [35] is an intensive dataset released by the Computational Physiology Laboratory of the Massachusetts Institute of Technology, which contains healthy data of patients in the intensive care unit of a large tertiary care hospital, including vital signs, medications, laboratory measurements, and survival data. The processed data are shown in Table 1.

5.2. Evaluation metrics

To better assess the model's performance, five kinds of evaluation metrics are chosen: DDI rate, Jaccard similarity, PRAUC, F1 score, and ROCAUC.

- **DDI rate.** For patient j, the DDI rate is calculated as follows:

$$DDI_j = \frac{\sum_{t=1}^{V_j} \sum_{k,l \in \{i:\hat{m}_{j,i}^{(t)}=1\}} 1\{D_{kl}=1\}}{\sum_{t=1}^{V_j} \sum_{k,l \in \{i:\hat{m}_{j,i}^{(t)}=1\}} 1}, \tag{20}$$

  where V_j is the total number of visits of patient j, $\hat{m}_{j,i}^{(t)}$ is the ith item of the drug combination recommended for patient j at the tth visit, D is the DDI adjacency matrix, and $1\{\cdot\}$ refers to the output of 1 when the expression in $\{\cdot\}$ is true, and 0 otherwise.

- **Jaccard similarity.** For patient j at the tth visit, the Jaccard similarity is calculated as follows:

$$Jaccard_j^{(t)} = \frac{|\{i : m_{j,i}^{(t)}=1\} \bigcap \{i : \hat{o}_{j,i}^{(t)}=1\}|}{|\{i : m_{j,i}^{(t)}=1\} \bigcup \{i : \hat{o}_{j,i}^{(t)}=1\}|}, \tag{21}$$

  where $m_j^{(t)}$ represents the true drug combination for patient j at the tth visit, $\hat{o}_{j,i}^{(t)}$ denotes the predicted drug combination for patient j at the tth visit, i represents the ith element in the combination, $|\cdot|$ denotes the cardinality operation, $\bigcap$ represents the intersection, and $\bigcup$ represents the union. After obtaining the Jaccard scores for each visit of patient j, the average of the Jaccard similarity scores for patient j is calculated as follows:

$$Jaccard_j = \frac{1}{V_j} \sum_{i=1}^{V_j} Jaccard_j^{(i)}, \tag{22}$$

  where V_j denotes the total number of visits of patient j.

- **F1 score.** The F1 score for patient j at visit t is the harmonic mean of precision and recall, calculated as follows:

$$Precision_j^{(t)} = \frac{|\{i : m_{j,i}^{(t)}=1\} \bigcap \{i : \hat{o}_{j,i}^{(t)}=1\}|}{|\{i : \hat{o}_{j,i}^{(t)}=1\}|}, \tag{23}$$

$$Recall_j^{(t)} = \frac{|\{i : m_{j,i}^{(t)}=1\} \bigcap \{i : \hat{o}_{j,i}^{(t)}=1\}|}{|\{i : m_{j,i}^{(t)}=1\}|}, \tag{24}$$

$$F1_j^{(t)} = \frac{2}{\frac{1}{Precision_j^{(t)}} + \frac{1}{Recall_j^{(t)}}}, \tag{25}$$

  Similarly, a patient's F1 score is obtained by taking the average of all visits for this patient:

$$F1_j = \frac{1}{V_j} \sum_{i=1}^{V_j} F1_j^{(i)}. \tag{26}$$

- **PRAUC.** To compute the PRAUC, a drug recommendation is regarded as an information retrieval task. For patient j at the tth visit:

$$PRAUC_j^{(t)} = \sum_{k=1}^{|\mathcal{M}|} Precision(k)_j^{(t)} \Delta Recall(k)_j^{(t)}, \tag{27}$$

$$\Delta Recall(k)_j^{(t)} = Recall(k)_j^{(t)} - Recall(k-1)_j^{(t)}, \tag{28}$$

  where k is the drug in the retrieved drug sequence. $|\mathcal{M}|$ is the total number of drugs. $Precision(k)_j^{(t)}$ is the precision at k in the retrieved list, and $\Delta Recall(k)_j^{(t)}$ is the change in the recall rate of a drug from drug $k-1$ to drug k. A patient's PRAUC is obtained by taking the mean value of all visits for this patient:

$$PRAUC_j = \frac{1}{V_j} \sum_{i=1}^{V_j} PRAUC_j^{(i)}. \tag{29}$$

- **ROCAUC.** To calculate ROCAUC, a drug recommendation is regarded as a binary classification problem. In this paper, samples with a true label of 1 are considered positive samples and samples with a true label of 0 are considered negative samples. First, the recommended drug combinations are sorted in descending order according to the size of the predicted value. Then, the drug with the largest predicted value is ranked $|\mathcal{M}|$ and the drug with the second largest predicted value is ranked $|\mathcal{M}| - 1$. The ROCAUC value is to add the rankings of all positive samples and subtract the number of combinations of two positive samples. For the tth visit of patient j,

$$ROCAUC_j^{(t)} = \frac{\sum_{positive} rank_k - \frac{P_j^{(t)}(1+P_j^{(t)})}{2}}{P_j^{(t)} N_j^{(t)}}, \tag{30}$$

Table 2
Performance comparison (true DDI rate of 0.0808 for the MIMIC-III dataset).

Model	DDI rate	Jaccard	PRAUC	F1-score	ROCAUC	Avg.# of Drugs
LR	0.0829 ± 0.0009	0.4865 ± 0.0021	0.7509 ± 0.0018	0.6434 ± 0.0019	0.9343 ± 0.0006	16.1773 ± 0.0942
ECC	0.0846 ± 0.0018	0.4996 ± 0.0049	0.6844 ± 0.0038	0.6569 ± 0.0044	0.9281 ± 0.0014	18.0722 ± 0.1914
RETAIN	0.0835 ± 0.0020	0.4887 ± 0.0028	0.7556 ± 0.0033	0.6481 ± 0.0027	0.9257 ± 0.0011	20.4051 ± 0.2832
LEAP	0.0731 ± 0.0008	0.4521 ± 0.0024	0.6549 ± 0.0033	0.6138 ± 0.0026	0.9172 ± 0.0014	18.7138 ± 0.0666
DMNC	0.0842 ± 0.0011	0.4864 ± 0.0025	0.7580 ± 0.0039	0.6529 ± 0.0030	0.9226 ± 0.0043	20.0000 ± 0.0000
GAMENet	0.0864 ± 0.0006	0.5067 ± 0.0025	0.7631 ± 0.0030	0.6626 ± 0.0025	0.9288 ± 0.0064	27.2145 ± 0.1141
SafeDrug	0.0617 ± 0.0005	0.5213 ± 0.0030	0.7647 ± 0.0025	0.6768 ± 0.0027	0.9344 ± 0.0009	19.9178 ± 0.1604
4SDrug	0.0600 ± 0.0004	0.5041 ± 0.0016	0.7611 ± 0.0026	0.6581 ± 0.0016	0.9351 ± 0.0013	17.5040 ± 0.0533
FFBDNet	0.0717 ± 0.0016	0.5292 ± 0.0020	0.7777 ± 0.0010	0.6833 ± 0.0017	0.9384 ± 0.0009	19.6900 ± 0.3000
MoleRec	0.0724 ± 0.0008	0.5305 ± 0.0033	0.7736 ± 0.0027	0.6843 ± 0.0029	0.9375 ± 0.0009	21.0893 ± 0.1788
RASNet w/o Recurrent Aggregation	0.0603 ± 0.0007	0.5302 ± 0.0021	0.7771 ± 0.0023	0.6840 ± 0.0021	0.9386 ± 0.0006	20.7050 ± 0.1613
RASNet w/o Drug Representation Module	0.0613 ± 0.0010	0.5353 ± 0.0026	0.7790 ± 0.0021	0.6890 ± 0.0025	0.9392 ± 0.0009	20.9234 ± 0.1582
RASNet w/o Exponential Controller	0.0802 ± 0.0005	0.5378 ± 0.0026	0.7814 ± 0.0028	0.6896 ± 0.0023	0.9404 ± 0.0009	21.8226 ± 0.1407
RASNet	**0.0599 ± 0.0009**	**0.5401 ± 0.0021**	**0.7882 ± 0.0025**	**0.6931 ± 0.0019**	**0.9418 ± 0.0007**	21.0434 ± 0.1466

where $P_j^{(t)}$ is the number of positive samples of patient j at the tth visit. $N_j^{(t)}$ is the number of negative samples of patient j at the tth visit. $rank_k$ is the ranking of the kth drug. The ROCAUC of each patient is averaged across all visits for that patient.

$$ROCAUC_j = \frac{1}{V_j} \sum_{i=1}^{V_j} ROCAUC_j^{(i)}. \tag{31}$$

5.3. Baselines

To evaluate its performance, RASNet is compared with the recent competitive state-of-the-art baseline methods, including the following methods:

- **LR**, which is logistic regression with L2 regularization.
- **ECC** [36], which is an integrated model for multi-label classification based on an SVM classifier.
- **RETAIN** [27], which is a clinically interpretable, two-level neural attention model designed to identify crucial past visits and significant clinical variables within those visits.
- **LEAP** [9], which is based on LSTM to capture label-dependent drug recommendations for patients.
- **DMNC** [37], which proposes an architecture for asynchronous two-view sequential learning to enhance patient encoding.
- **GAMENet** [34], which presents a graph-augmented memory network that embedded multiple knowledge graphs using a fusion mechanism-based GCN into the memory component and enabled attention-based memory search using a query generated from longitudinal patient records.
- **SafeDrug** [10], which is a DDI-controllable drug recommendation model that uses the molecular structures of drugs and effectively models DDIs.
- **4SDrug** [32], which is a symptom-based set-to-set small and safe drug recommendation model.
- **MoleRec** [33], which is a molecular substructure-based approach used to recommend drugs by modeling the relationship of drug substructures and the patient's health condition.
- **FFBDNet** [16], which is a feature fusion and bipartite decision network developed for drug recommendation, aiming to integrate external medical knowledge and enhance the effectiveness of recommendations.

5.4. Implementation details

The dataset is divided into training, validation, and test data sets, with a ratio of $\frac{2}{3} : \frac{1}{6} : \frac{1}{6}$. δ is set to 2. All of the following experiments use Python 3.8 and PyTorch 2.0.1. Experiments are conducted on a computer with a GeForce RTX 3090 GPU. During the training process, the learning rate is set to 5×10^{-4} and the training epochs is set to 50. The Adam optimizer [38] is used for model training. It is worth noting that all baseline models are trained and implemented according to their original papers with optimal parameters.

6. Results and discussion

6.1. Performance comparison

Table 2 provides a comparative analysis of various models. The performance of the best model is represented by the bold font, and the performance of the second best model is represented by underlining. The results reveal that longitudinal models exhibit superior performance, attributed to their ability to capture the patient's historical health more comprehensively. In DDI evaluation, the smaller the DDI value is, the higher is the safety of the drug combination recommended by the model. SafeDrug stands out as a top performer because it uses a distinctive method to encode drug information. This method relies not only on molecular structures but also on the regulation of DDI loss through an adaptive loss controller. GAMENet exhibits high dependency on historical combinations and results in a relatively high DDI. RASNet exhibits the lowest DDI rate among the compared baselines, indicating a superior safety of drug combination. 4SDrug aims to minimize the size of suggested drug combinations, and it is observed that it achieves the minimum number of recommended drugs at the cost of decreased recommendation accuracy. FFBDNet uses attention to capture useful information from historical visits and combines it with external drug information to enable safe and effective drug recommendations. MoleRec models the interactions among drugs through their molecular substructures and discovers the connection between patient's health condition and drugs' substructures. FFBDNet and MoleRec show good performance on the MIMIC-III dataset but are still less effective than RASNet. This study selects SafeDrug, FFBDNet, and MoleRec as the top-performing baseline models to conduct a comparative analysis of performance across different visits. Fig. 3 shows the performance of each model across different visits. FFBDNet is particularly effective in capturing the patient history information for patients with multiple visits, as demonstrated by its superior performance. By contrast, SafeDrug and MoleRec both leverage recurrent neural networks (RNNs) to model patient history, resulting in a smoother performance. RASNet leverages the benefits of RNNs to effectively mitigate noise in historical data, thereby achieving the highest performance among the compared methods. Compared with baseline models, RASNet exhibits superior prediction accuracy. The experimental findings demonstrate that RASNet, equipped with the recurrent aggregation module, outperforms existing methods in terms of recommending safer and more effective drug combinations. Furthermore, the exponential controller module in RASNet effectively enhances the balance between accuracy and safety compared with contemporary approaches.

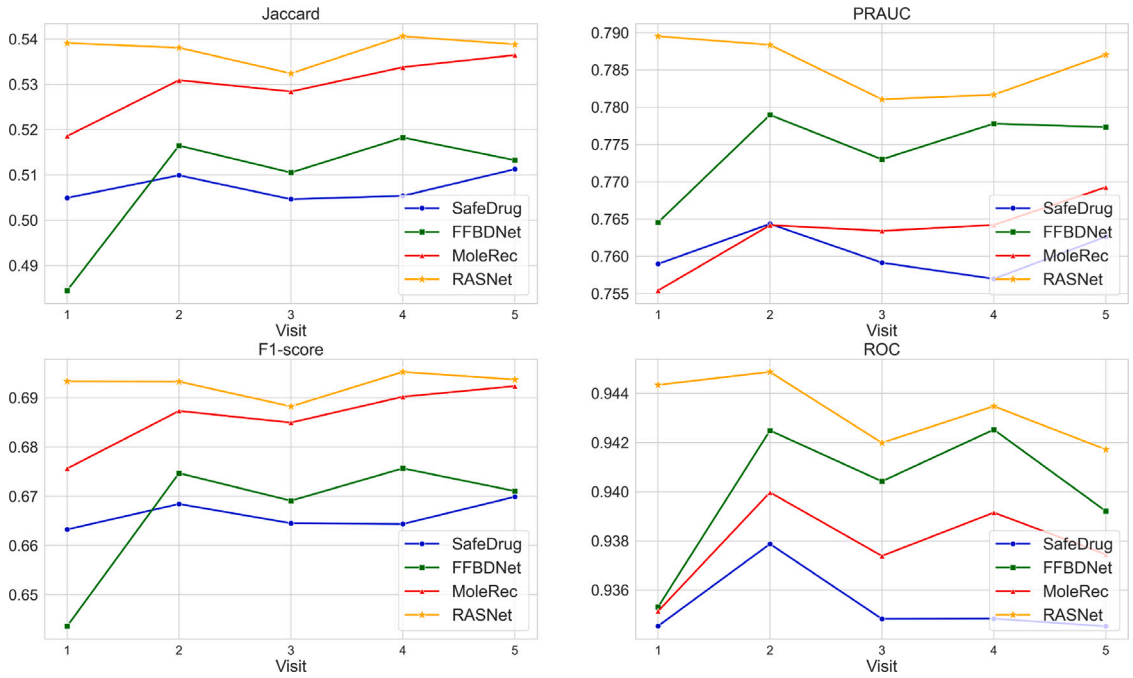


Fig. 3. The performance of different models across multiple visits.

6.2. Ablation study

To evaluate the individual contributions of RASNet's modules, this study conducts comprehensive ablation studies by systematically excluding specific components. These experiments facilitate an in-depth analysis of the influence of each module on RASNet's overall performance. The paper outlines three distinct ablation experiments for a thorough examination:

- **RASNet** w/o **recurrent aggregation.** The primary function of this module is to introduce similarity comparisons among historical health conditions to enhance the model's capacity for personalized history modeling. In particular, chronic diseases with periodic patterns may lead to inaccurate diagnoses on certain occasions, and models based on these diagnostic results need to consider the influence of this noisy data. In this ablation experiment, a noteworthy decrease in accuracy with RASNet is observed. The exclusion of the recurrent aggregation network module significantly compromises the performance of RASNet. Therefore, RASNet could eliminate noisy data from the historical health information, enabling a more precise representation of the patient.
- **RASNet** w/o **drug representation module.** The exclusion of the drug representation module also led to a noteworthy decrease in RASNet's performance. This module plays a pivotal role in transforming drugs into vector representations, facilitating their processing within the model. Its absence hampers the model's ability to effectively represent drugs, consequently causing a loss in precision capabilities for drug recommendations. As a result, the findings from this ablation experiment underscore the critical significance of the drug representation module in enhancing the drug recommendation effectiveness of RASNet.
- **RASNet** w/o **exponential controller.** Even in the absence of the exponential controller, where β is set to zero, RASNet consistently demonstrates robust performance in drug recommendation. The primary role of the exponential controller is to fine-tune the weight of the DDI loss, balancing recommendation accuracy and DDI rate. Although the model's adherence to DDI constraints may diminish with the removal of β, it still manages to maintain a high level of overall recommendation accuracy. This observation underscores RASNet's robustness as it could sustain strong performance across varying DDI constraint conditions.

Table 3
The comparison of model efficiency.

Model	# of param	Training time/epoch	Inference
RETAIN	285 489	7.15 s	1.64 s
LEAP	433 286	102.42 s	9.97 s
DMNC	521 866	1225.98 s	64.61 s
GAMENet	444 209	59.92 s	8.64 s
SafeDrug	388 798	53.55 s	4.25 s
4SDrug	79 681	13.55 s	4.25 s
MoleRec	507 060	305.91 s	32.97 s
FFBDNet	528 898	51.22 s	4.60 s
RASNet	329 797	24.43 s	3.38 s

By comparing the results of the ablation experiments, **RASNet** w/o **recurrent aggregation** exhibits the poorest performance, which shows that the recurrent aggregation network could improve the performance of the model and effectively capture the patient's representation. By comparing the **RASNet** w/o **drug representation module** and **RASNet**, it is shown that the DDI and EHR diagrams in the drug graph encoding module could learn a more comprehensive drug representation and improve the performance of the model. This study reveals that in the absence of the DDI loss constraint, the **RASNet** w/o **exponential controller** exhibits relatively high scores across all evaluation metrics. However, its DDI rate significantly surpasses that of RASNet, indicating the effectiveness of the exponential controller in simultaneously preserving model performance and minimizing DDIs. In summary, RASNet exhibits a clear advantage in both safety and accuracy compared with all of its variants, highlighting its performance in drug recommendation.

6.3. Model efficiency

Table 3 provides a comparison of the time complexity of different models. To ensure a fair comparison, the embedding dimensions are consistent across all models. RASNet exhibits relatively lower time

Table 4
Analysis of recurrent aggregation parameter φ.

φ	DDI rate	Jaccard	PRAUC	F1-score	ROCAUC	# of med.
0.01	0.0622	0.5354	0.7796	0.6882	0.9402	20.88
0.05	0.0599	0.5401	0.7882	0.6931	0.9418	21.04
0.1	0.0612	0.5355	0.7801	0.6889	0.9405	21.03
0.5	0.0609	0.5335	0.7785	0.6866	0.9393	21.09
1	0.0603	0.5302	0.7771	0.6840	0.9387	20.70

Table 5
Analysis of exponential controller hyper-parameter ϕ.

ϕ	DDI rate	# of Med.	Jaccard	PRAUC	F1-score	ROCAUC
0	0.0732	21.67	0.5392	0.7837	0.6917	0.9417
1	0.0599	21.04	0.5401	0.7882	0.6931	0.9414
2	0.0473	19.83	0.5309	0.7809	0.6852	0.9397
3	0.0279	18.30	0.5162	0.7700	0.6724	0.9371
4	0.0150	17.09	0.4975	0.7601	0.6556	0.9328
5	0.0056	16.12	0.4750	0.7430	0.6356	0.9262
6	0.0007	15.27	0.4564	0.7272	0.6183	0.9184

Table 6
Comparison of different DDI rate.

Model	DDI rate	Jaccard	PRAUC	F1-score	ROCAUC
SafeDrug($\gamma = 0$)	0.0073	0.4198	0.6902	0.5743	0.8924
FFBDNet(greedy)	0.0019	0.4361	0.7061	0.5978	0.8834
MoleRec($\gamma = 0.002$)	0.0183	0.4252	0.6570	0.5887	0.8632
RASNet($\phi = 6$)	**0.0007**	**0.4564**	**0.7272**	**0.6183**	**0.9184**

and space complexities. LEAP [9] and DMNC [37] used sequential recommendation, which entailed recommending drugs one by one, thereby increasing both time and space complexities. GAMENet [34] introduced a graph-enhanced memory module, significantly augmenting the model's space complexity. SafeDrug excelled in efficiency by focusing on the relationship between drug molecules and individual patient health profiles. By contrast, 4SDrug [32] used a simple set-to-set recommendation approach, avoiding intricate neural architectures, which naturally facilitates efficient minibatch training, thus optimizing efficiency. Conversely, MoleRec [33] introduces numerous substructures and computes their dependencies, leading to increased spatial and temporal complexity. FFBDNet [16] uses attention-based feature fusion and a large external medication representation module, leading to a high space complexity. By contrast, RASNet uses recurrent aggregation for the personalized historical modeling of patient information, effectively reducing the number of parameters and computational load. During the drug encoding phase, a smaller external drug module and an attention mechanism are used to achieve safer and more precise drug representations. Consequently, RASNet achieves superior accuracy with relatively lower space and time complexity.

6.4. Parameter analysis

- **Analysis of recurrent aggregation parameter φ.** Table 4 presents a comparative analysis of the model's performance under varying parameter values for φ. The results offer intriguing insights. Notably, this study observes that thoughtful adjustments to the φ parameter within a specific range lead to a substantial improvement in the model's performance. The optimal performance is achieved when $\varphi = 0.05$, with larger values of φ retaining more noise and smaller values providing less historically valid information. Although changes in the φ value may have some impact on other performance metrics such as Jaccard or F1 scores, it appears that the DDI rate remains relatively unaffected. Empirical evaluations indicate that as the value of φ increases, more noise is retained, whereas a smaller φ leads to the loss of valuable information. To balance safety and accuracy, this study selects φ with a value of 0.05, optimizing the trade-off.

Table 7
Denoise experiments.

Model	Denoise	DDI	Jaccard	PRAUC	F1-score	ROCAUC
SafeDrug	No	0.0589	0.4877	0.7879	0.6528	0.9401
	Yes	0.0551	0.5626	0.8160	0.7190	0.9460
FFBDNet	No	0.0770	0.5406	0.8446	0.6986	0.9517
	Yes	0.0762	0.5468	0.8377	0.7022	0.9488
MoleRec	No	0.0595	0.6449	0.8532	0.7833	0.9534
	Yes	0.0727	0.6807	0.8646	0.8084	0.9552

- **Analysis of exponential controller hyperparameter ϕ.** In this study, the DDI rate and various performance metrics are evaluated for different weight settings, and the results are detailed in Table 5. The table lists the average results of five experiments after reaching convergence. When the parameter ϕ is started to be tuned, the DDI rate significantly decreases by 1.33% and the other metrics remain relatively stable. This observation suggests that the loss function of RASNet pays great attention not only to the safety of drug combinations but also to the recommendation accuracy.
- **Analysis of different DDI rate.** In Table 6, this paper compares the results of different models after minimizing the DDI rate. Among them, SafeDrug ($\gamma = 0$) sets the threshold to 0. FFBDNet(using a greedy strategy) aims to minimize DDI by adopting the greedy approach. However, the peculiarity of MoleRec's loss function precludes setting the threshold to zero. This study uses a threshold of 0.002 for comparative purposes. The experimental findings demonstrate RASNet's outstanding performance, particularly in Jaccard index and other metrics.

6.5. Denoise study

An experimental denoise analysis is conducted to assess the effectiveness of the recurrent aggregation module proposed in this paper. The test dataset is filtered for patients exhibiting periodic variations across all visits, and such patients could be detected using the recurrent aggregation module.

This study selects three case studies, each consisting of three visits. The first and third visits of the patients exhibit a specific disease, whereas the second visit remains unaffected. A comparative analysis is then conducted, comparing screened patients (excluding the second visit) with those unprocessed (retaining all historical visits). Table 7 presents the outcomes for SafeDrug, FFBDNet, and MoleRec. The "Denoise" column in Table 7 indicates whether RASNet removes redundant or invalid visits from historical diagnoses; "Yes" signifies processed patients, and "No" denotes unprocessed ones. The experimental results of SafeDrug and MoleRec indicate improvements across various metrics after processing. FFBDNet, which uses RNNs with an attention mechanism for patient encoding, exhibits a minor decrease in PRAUC and ROCAUC. FFBDNet will benefit from training efficiency because of the reduced samples of multiple visits after processing.

6.6. Case study

In this case study, a patient who visited the hospital five times is selected from the test set and its drug combination is predicted by SafeDrug, FFBDNet, and MoleRec. The detailed information related to the drug recommendation for this patient across different models is presented in Table 8. To facilitate the comparison, a concise and visual representation is shown in Fig. 4. As shown in Fig. 4 and Table 8, "Correct" indicates the number of correctly recommended drugs, "Missed" indicates the drugs present in actual prescriptions but not predicted by the model, and "Unseen" represents the drugs predicted by the model but absent in real prescriptions. To highlight

Table 8
Recommended drugs for a patient with multiple visits.

Visit	Model	Drug combination	Correct\Missed\Unseen
1	Ground Truth	'A12C', 'A02B', 'A04A', 'N06A', 'C03C', 'J01D', 'A06A', 'C03D', 'A02A', 'N03A', 'C07A', 'N02B', 'A07A', 'H01C', 'C10A', 'C01C', 'A05A'	17
	SafeDrug	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'A12B', 'N02A', 'J01M', 'N05C', 'J01D', 'A04A', 'B03B', 'H01C'	8\9\6
	FFBDNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'A12B', 'N02A', 'H01C'	6\11\3
	MoleRec	'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'A12B', 'N02A', 'J01M', 'N05C', 'B02B', 'C03D', 'H01C', 'A09A'	6\11\7
	RASNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'N02A', 'J01M', 'N05C', 'J01D', 'B02B', 'A04A', 'H01C'	8\9\5
2	Ground Truth	'B05C', 'A12C', 'N02B', 'A12A', 'A02B', 'A04A', 'N02A', 'N06A', 'N05B', 'C03C', 'J01D', 'A06A', 'C03D', 'C01D', 'N03A', 'C07A', 'N05C', 'A07A', 'H01C', 'C10A', 'A05A'	21
	SafeDrug	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'J01M', 'N05C', 'J01D', 'A04A', 'B03B', 'H01C'	14\7\3
	FFBDNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'N02A', 'N06A', 'J01D', 'A04A', 'B03B', 'H01C'	11\10\1
	MoleRec	'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'C03C', 'A12B', 'N02A', 'J01M', 'N05C', 'J01D', 'B02B', 'A04A', 'C03D', 'H01C', 'A09A'	12\9\4
	RASNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'C07A', 'C03C', 'N02A', 'J01M', 'N05C', 'J01D', 'B02B', 'A04A', 'C03D', 'H01C'	13\8\2
3	Ground Truth	'B05C', 'A12C', 'A07A', 'C03C', 'A12A', 'A02B', 'B01A', 'A06A', 'A04A', 'N02A', 'N06A', 'N05A', 'N03A', 'J01C', 'C01B', 'N05C', 'N02B', 'H01C', 'C10A', 'C01C', 'N05B'	21
	SafeDrug	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C03C', 'A12B', 'N06A', 'J01M', 'B01A', 'J01D', 'N03A', 'A04A', 'B03B', 'N05B', 'C03D', 'H01C'	13\8\6
	FFBDNet	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'C01C', 'A07A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'B01A', 'J01D', 'N03A', 'A04A'	13\8\4
	MoleRec	'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'C01C', 'A07A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'J01M', 'J01D', 'A04A', 'C03D', 'H01C', 'A09A'	11\10\7
	RASNet	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12A', 'A12C', 'A07A', 'C07A', 'C03C', 'N02A', 'N06A', 'B01A', 'J01D', 'N05A', 'N03A', 'A04A', 'C03D', 'H01C', 'N05B'	16\5\4
4	Ground Truth	'B05C', 'A12C', 'N01A', 'A02B', 'A04A', 'N05A', 'J01D', 'N02A', 'A06A', 'A01A', 'C03D', 'C07A', 'A07A', 'H01C', 'C03C'	15
	SafeDrug	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'N01A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'J01M', 'C01B', 'N05C', 'J01D', 'A04A', 'B03B', 'H01C'	13\2\7
	FFBDNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'N01A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'B01A', 'N05C', 'J01D', 'N03A', 'B03B', 'H01C'	11\4\7
	MoleRec	'A01A', 'A02B', 'A06A', 'B05C', 'A12A', 'A12C', 'A07A', 'N01A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'J01M', 'N05C', 'J01D', 'A04A', 'C03D', 'H01C', 'A09A'	14\1\6
	RASNet	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'N01A', 'C07A', 'C03C', 'N02A', 'J01M', 'B01A', 'N05C', 'J01D', 'N05A', 'A04A', 'C03D', 'H01C'	15\0\4
5	Ground Truth	'A12C', 'N01A', 'C03C', 'A02B', 'A02A', 'B01A', 'A01A', 'A04A', 'N05A', 'N06A', 'J01M', 'A03F', 'J01D', 'N02A', 'A06A', 'C03D', 'A07E', 'C01B', 'D07A', 'N03A', 'C07A', 'M04A', 'A07A', 'H01C', 'C10A', 'A05A'	26
	SafeDrug	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C03C', 'A12B', 'N06A', 'J01M', 'C01B', 'H03A', 'J01D', 'N05A', 'A04A', 'B03B', 'N05B', 'H01C'	12\14\6
	FFBDNet	'N02B', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'H03A', 'J01D', 'A04A', 'B03B', 'C03D', 'H01C'	12\14\5
	MoleRec	'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C07A', 'C03C', 'A12B', 'N02A', 'N06A', 'J01M', 'N05C', 'J01D', 'B02B', 'A04A', 'C03D', 'H01C', 'A09A'	13\13\5
	RASNet	'N02B', 'A01A', 'A02B', 'A06A', 'A12C', 'A07A', 'C07A', 'C03C', 'N02A', 'J01M', 'B01A', 'H03A', 'J01D', 'B02B', 'N05A', 'A04A', 'A05A', 'C03D', 'H01C'	16\10\3

the best model, this study marks the model with the largest number of correctly predicted drugs and the smallest DDI value drug combination in red color. By comparing the "Correct" and "Unseen" categories for each model, it could be seen that the RASNet outperforms the other models in terms of recommendation accuracy. In particular, RASNet excels in accurate drug prediction while minimizing the occurrence of redundant recommendation. This indicates that RASNet could provide more accurate predictions of the drugs that a patient truly needs while avoiding unnecessary drug recommendation.

A more in-depth analysis of this case study underscores the significant advantage of the RASNet model in terms of personalized drug recommendations based on a patient's historical medical visit records. Through the application of the recurrent aggregation method, RASNet effectively captures information similar to the current visit, thereby enhancing the patient's ability to create personalized historical records. Furthermore, the exponential controller mechanism adeptly manages DDIs, maintaining a lower DDI rate while ensuring the model's exceptional performance in drug recommendation accuracy. The case study further demonstrates the effectiveness and superiority of the RASNet model in the field of drug recommendation. RASNet enhances patient representation by reducing data noise and comprehensively addresses DDI, resulting in more accurate and safer drug recommendation. Therefore, RASNet may help clinicians and patients make reasonable drug treatment decisions.

6.7. Discussion

This study seeks to address the research gap in medication recommendation, recognizing the limitations of existing methods in managing patients' condition fluctuations. For instance, although a patient's electronic medical records may initially list anemia due to end-stage renal disease (285.21), subsequent visits may not exhibit this condition,

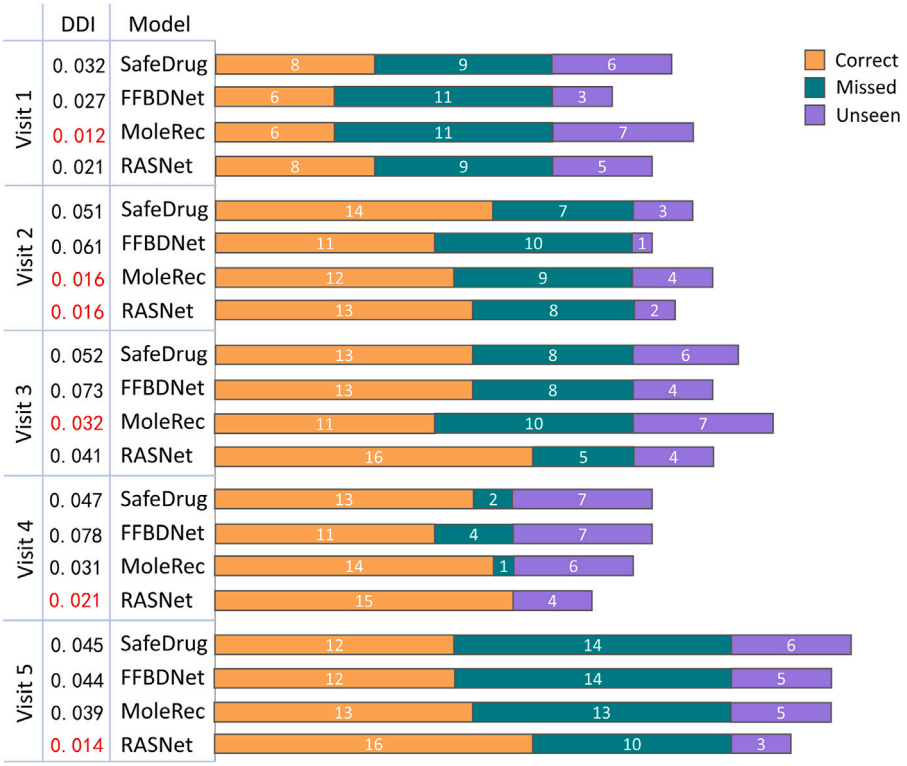


Fig. 4. Drug recommendation for a patient with 5 visits.

indicating temporal disease occurrences. This highlights the need to consider such periodic changes, which may affect recommendation accuracy. Concurrently, drug recommendation should consider not only drug efficacy but also the potential DDIs because the safety in combination therapy is a critical research which will minimize adverse effects and enhance treatment outcomes.

This study develops a recurrent aggregation network to address the influence of periodic variations. The network filters redundant visits by integrating and comparing visit data with the current visit. Moreover, an exponential controller is introduced to ensure the safety and precision of drug combinations. The controller dynamically adjusts the DDI loss, preventing excessive reduction in prediction loss, thus maintaining recommendation accuracy. Experimental outcomes demonstrate significant enhancements in DDI, Jaccard, PRAUC, F1, ROCAUC, and model efficiency for RASNet.

Although the study presents valuable insights, its limitations should not be overlooked. First, RASNet solely focuses on a patient's historical data from the current visit, neglecting the potential impact of inter-patient relationships. Its efficacy may be compromised when dealing with large datasets where each visit contains numerous patients. In addition, the model's reliance on diagnosis, procedure, and medication data as auxiliary information to predict drugs excludes other relevant factors such as patient demographics and laboratory test results.

7. Conclusion

The field of drug recommendation strives to leverage patients' EHRs to provide effective and safe medication combinations. Nevertheless, it continues to grapple with challenges related to inadequate patient information modeling and the complex control of DDIs. In response to these challenges, this paper presents a novel approach called RASNet that is based on a recurrent aggregation neural network for safe and efficient drug recommendation. RASNet enhances patient representation by identifying historical states that are closely similar to the patient's current condition. Furthermore, RASNet introduces an exponential controller to ensure the safety and precision of recommended drug combinations. The evaluation of RASNet on the MIMIC-III dataset has demonstrated its superior efficiency, accuracy, and safety. In future research, the following areas will be explored: (i) incorporating medical ontology to improve the model's comprehension of relationships among medical concepts and (ii) integrating drug time-dependency considerations using a patient's medication history to enhance the precision of drug recommendation. These strategic enhancements have the potential to improve both the performance and safety of drug recommendations.

CRediT authorship contribution statement

Qiang Zhu: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Conceptualization. **Feng Han:** Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Huali Yang:** Visualization, Writing – review & editing, Validation. **Junping Liu:** Validation, Data curation. **Xinrong Hu:** Writing – review & editing. **Bangchao Wang:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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